



Systemic Risk in Multiplex Financial Networks

Centrality, Contagion, and Policy Intervention

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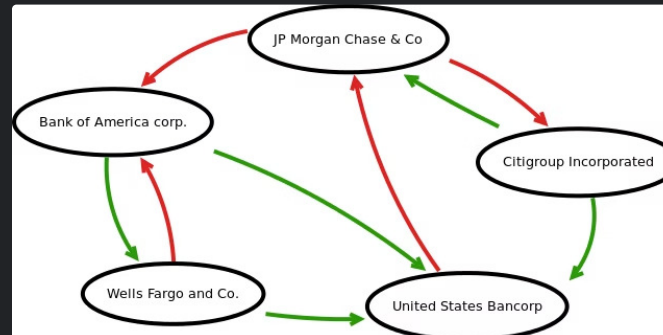
Finance into Graph Theory

The Financial Reality

- **Concept:** Interbank Lending & Credit Risk
- **Example:** Bank i lends \$10 Million to Bank j .
- **The Dynamic:**
 - Money flows from Bank i to Bank j .
 - Risk Exposure points from Bank i to Bank j (Bank i is now exposed to Bank j 's default).

The Network Abstraction

- **Nodes (Vertices V):** Financial Institutions (e.g., Node i , Node j).
- **Directed Edges (E):** Credit Exposure. A directed edge $i \rightarrow j$ exists if node i holds a claim against node j .
- **Edge Weight (W_{ij}):** The dollar volume of the exposure (e.g., \$10M).
- **Adjacency Matrix (A):** The global banking system is compressed into an asymmetric matrix A , where $A_{ij} > 0$.



Core Research Questions



Metric

Measuring systemic importance beyond single layers. (Contribution: **MultiRank**)



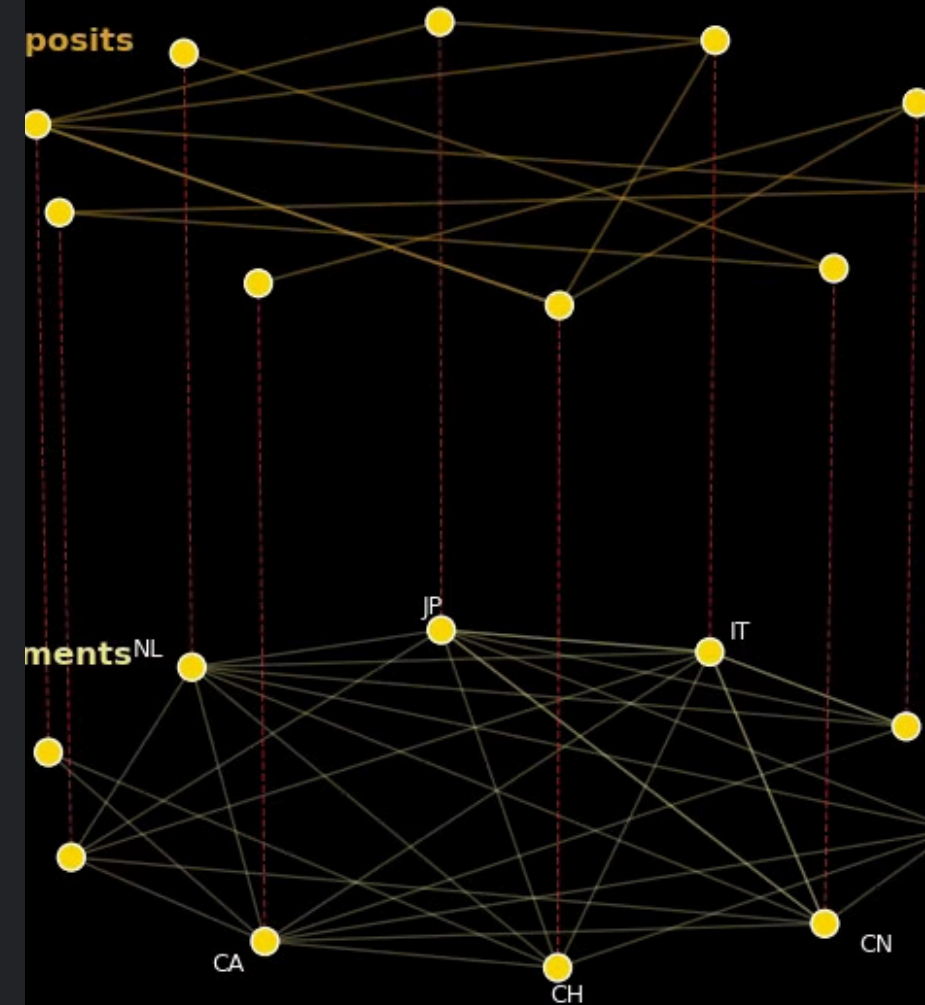
Dynamics

Unveiling the propagation mechanism of financial shocks.
(Contribution: **Multiplex Contagion: Solvency & Liquidity Contagion Model**)



Policy

Optimizing intervention strategies: Targeting the **Source (Patient Zero)** vs. the **Core (Hub)**?



The Foundation: Single-Layer PageRank

Scalar Definition & Core Concepts

$$x_i = \mu \sum_{j=1}^N \frac{a_{ji}}{k_j^{\text{out}}} x_j + w, \quad i = 1, 2, \dots, N$$

- **Adjacency Matrix:** $a = (a_{ij})$
- **Out-degree:** $k_j^{\text{out}} = \max(K_j^{\text{out}}, 1)$, where $K_j^{\text{out}} = \sum_{i=1}^N a_{ji}$
- **Damping factor:** $\mu \in (0, 1)$

Explicit Definition of x

x_i represents the PageRank centrality of node i (the steady-state probability of finding a random surfer at node i). The vector x must satisfy the normalization condition:

$$\sum_{i=1}^N x_i = 1$$

Matrix Representation

Define the Random Walk Matrix P :

$$P_{ij}^{PR} = \frac{a_{ji}}{k_j^{\text{out}}}$$

Teleportation Term (w)

$$w = \frac{1}{N} \sum_{j=1}^N [(1 - \mu) + \mu \delta(K_j^{\text{out}}, 0)] x_j$$

Note: w handles dangling nodes and random jumps, ensuring the network is strongly connected.

Computing PageRank: Iteration & Eigensystem

Recursive Approximation (Power Iteration)

The PageRank vector x can be recursively approximated. This simulates the step-by-step transitions of a random surfer until convergence ($x^{(t+1)} \approx x^{(t)}$):

$$x^{(t+1)} = \mu P x^{(t)} + w$$

x as an Eigenvector (The Matrix)

To show that x is an eigenvector with an eigenvalue of 1, we absorb the teleportation term w into a new matrix. We define the Matrix G with elements:

$$G_{ij} = \mu P_{ij}^{PR} + \frac{1}{N} [(1 - \mu) + \mu \delta(K_j^{\text{out}}, 0)]$$

The PageRank equation simplifies to:

$$x_i = \sum_{j=1}^N G_{ij} x_j$$

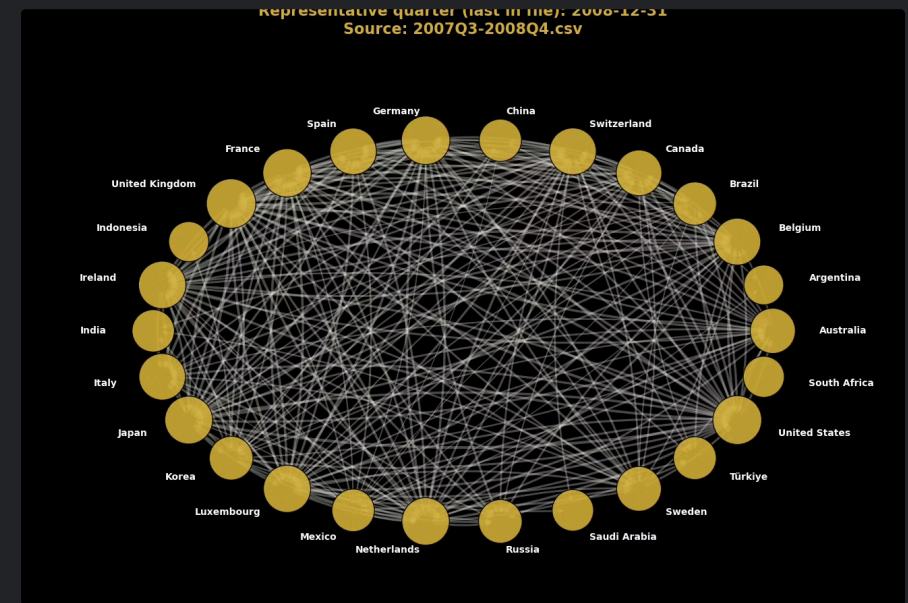
In vector form, this is exactly the eigenvalue equation for $\lambda = 1$:

$$x = Gx$$

Therefore, the PageRank vector x is the principal eigenvector of the Matrix G associated with eigenvalue $\lambda_1 = 1$.

Why Single-Layer Analysis Fails

- The 2008 financial crisis **revealed hidden interconnections** and cascading failures.
- Traditional systemic risk analysis often focuses solely on a '**Total Assets**' perspective.
- Financial reality is **multiplex**: loans, derivatives, and bonds operate on distinct, interconnected layers, not just a single aggregate.



MultiRank on a Multiplex Network

Step 1: Aggregating Layers into a 'Coloured Network'

Aggregates M layers using layer influence vector z . (Matrix representation)

$$C_{ij} = \sum_{\alpha=1}^M a_{ij}^{[\alpha]} z^{[\alpha]}$$

Layer Weights (B matrices)

$$B_{\alpha i}^{in} = \frac{\sum_{j=1}^N a_{ji}^{[\alpha]}}{W^{[\alpha]}}, \quad B_{\alpha i}^{out} = \frac{\sum_{j=1}^N a_{ij}^{[\alpha]}}{W^{[\alpha]}}$$

Total Weight (W)

$$W^{[\alpha]} = \sum_{i=1}^N \sum_{j=1}^N a_{ij}^{[\alpha]}$$

Step 2: The Generalized MultiRank Equation

$$x_i^{(t+1)} = \mu \sum_{j=1}^N \frac{C_{ji}}{K_j} x_j^{(t)} + w v_i, \quad \begin{cases} K_j = \max(1, \sum_{i=1}^N C_{ji}) \\ v_i = \theta \left(\sum_{j=1}^N (C_{ij} + C_{ji}) \right) \end{cases}$$

$$\text{Vector Form: } \mathbf{x}^{(t+1)} = \mu P^{MR} \mathbf{x}^{(t)} + w \mathbf{v}, \quad \text{where } P_{ij}^{MR} = \frac{C_{ji}}{K_j}$$

MultiRank Step 2: The Coupled System

The MultiRank algorithm finds node centrality x_i and layer influence $z^{[\alpha]}$ by solving a set of coupled equations:

Layer Influence

$$z^{[\alpha]} = \frac{1}{\mathcal{N}} \left[W^{[\alpha]} \right]^a \left[\sum_{i=1}^N B_{\alpha i}^{in} (x_i)^{s\gamma} \right]^s$$

$\mathcal{N}$ is a normalization constant.

Key Properties of the Coupled System:

$$\sum_{\alpha=1}^M z^{[\alpha]} = 1$$

The behavior of layer influence is tuned by three key parameters:

- Parameter $a \in \{0, 1\}$: Determines if layer influence scales with its total weight ($a = 1$) or is normalized ($a = 0$).
- Parameter $s \in \{-1, 1\}$: The strategic core.
 - If $s = 1$, layers gain influence by hosting many high-centrality nodes.
 - If $s = -1$, it rewards "elite" layers: influence increases if the layer is driven by a few highly influential nodes.
- Parameter $\gamma > 0$: Tunes the non-linear contribution of node centralities.

The Synthesis: Collapsing back to PageRank

How does the complex MultiRank system behave if we apply it to a single-layer network ($M = 1$)? It perfectly collapses into standard PageRank, but it requires specific parameter conditions.

The Boundary Conditions

We observe a single layer ($M = 1$) and remove any layer-size or node-centrality bias by setting $\alpha = 0$ and $\gamma = 0$. This neutralizes the layer influence, forcing $z^{[1]} = 1$.

The 4-Step Collapse (Conceptual)

01

Matrix Equivalence

Because $z^{[1]} = 1$, the coloured adjacency matrix reverts to the standard adjacency matrix ($C_{ij} = a_{ij}$).

02

Out-Strength Alignment

The generalized out-degree naturally simplifies to the standard out-degree ($K_j = k_j^{\text{out}}$), seamlessly handling dangling nodes just like PageRank.

03

Transition Probabilities

As a direct result of Steps 1 & 2, the MultiRank transition matrix becomes the standard random walk matrix ($P_{ij}^{MR} = P_{ij}^{PR}$).

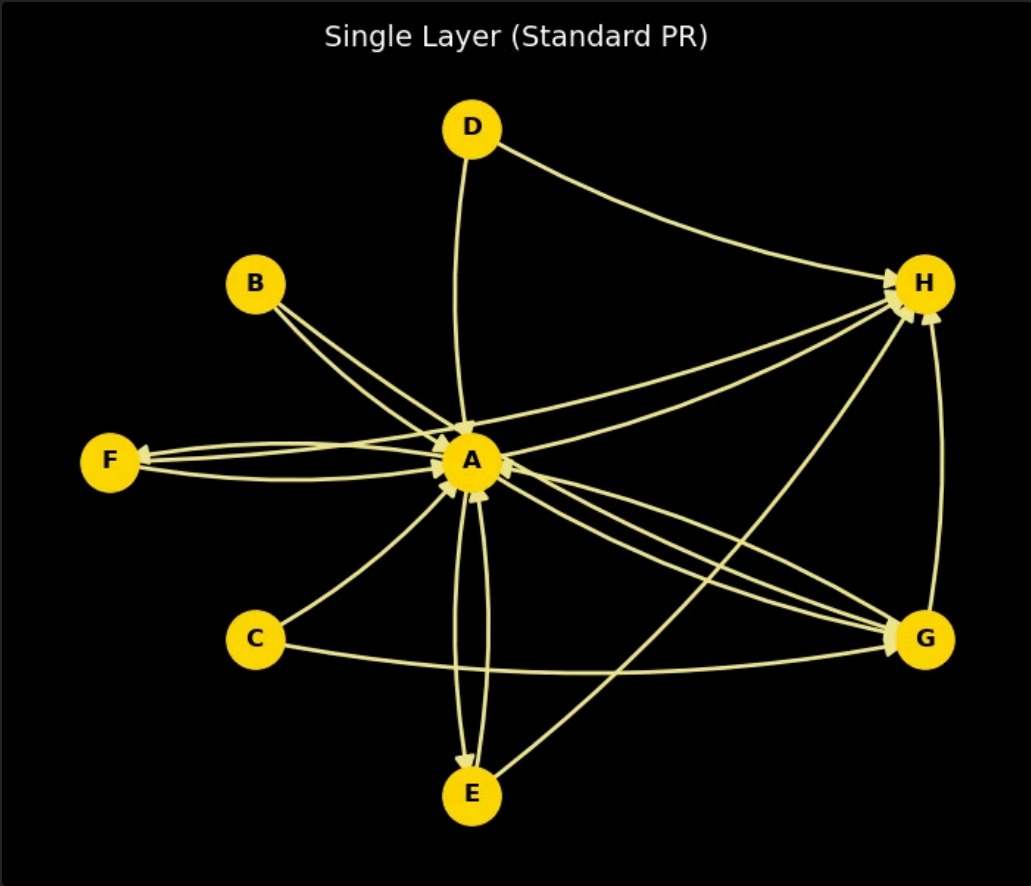
04

The Final Equation

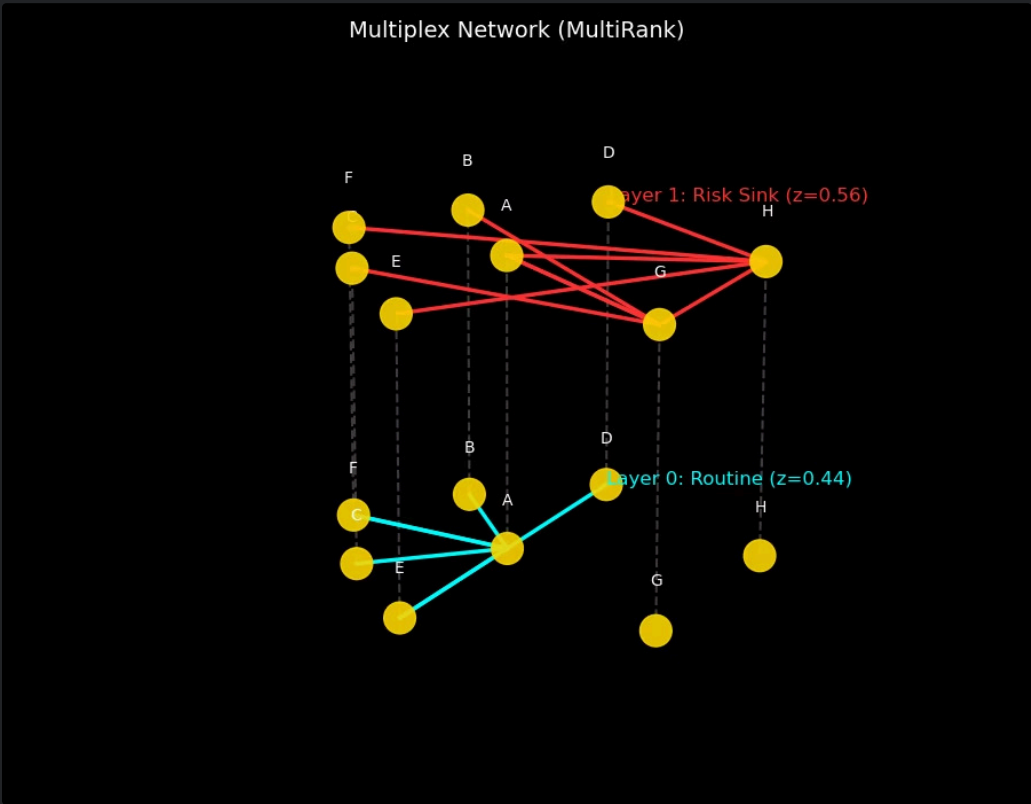
The coupled system completely decouples. The vector equation becomes $x = \mu P^{PR} x + (1 - \mu)v$.

Because the transition matrices are identical, the unique probability vector x solved by MultiRank precisely coincides with the PageRank centrality vector!

Why MultiRank? Unmasking the True Core



Single-Layer View: All connections look equal.



MultiRank View: Reveals a 'Trap' layer.

The Illusion vs. The Reality

Standard PageRank x_i

Node H: **0.26**

Node A: **0.25**

Verdict: PageRank is distracted by Node A's high volume of routine connections. It reveals A and H are almost equally important.

MultiRank x_i

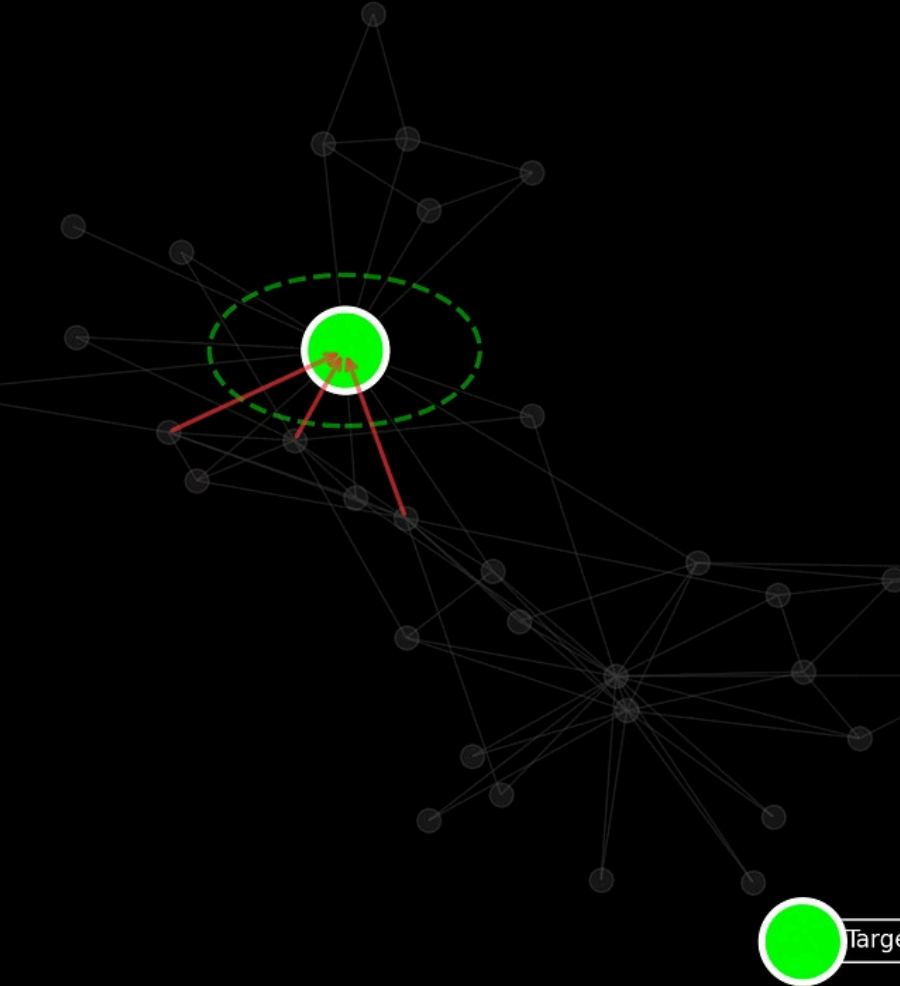
Node H: **0.34** (Massive Surge!)

Node A: **0.25**

Verdict: MultiRank dynamically assigns more weight to the "trap" layer ($z_1 = 0.56$ vs $z_0 = 0.43$). It reveals Node H is an undisputed black hole absorbing the network's influence.

❏ **Key Takeaway:** Standard algorithms count the *number* of connections. MultiRank evaluates the *context* of connections, preventing critical nodes from hiding behind complex structures.

TRANSITION: PRECISION INTERVENTION



The simulation
Model(Methodology)

Methodology: Data & Network Construction

Data Source

Bank for International Settlements (BIS) - Locational Banking Statistics.

Timeline Selection (The Crisis Arc)

- 2007 Q2: Pre-Crisis baseline.
- 2008 Q4: Peak of the Global Financial Crisis (Lehman collapse).
- 2009 Q4: Post-Crisis recovery phase.

Multiplex Topology

- Nodes: Reporting Countries
- **Layer 0 (Total Claims)**: Represents broad systemic exposure across all instruments.
- **Layer 1 (Loans & Deposits)**: Represents direct, highly liquid funding channels.

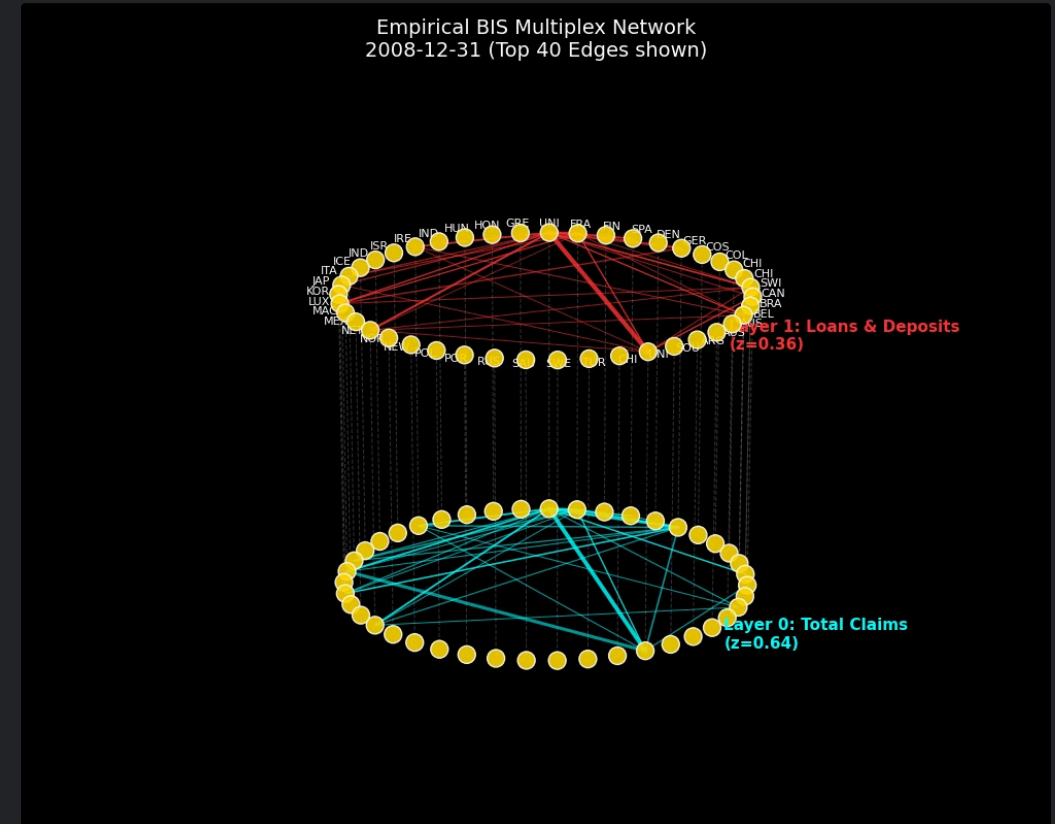


Fig: Snapshot of the empirical multiplex banking network.

Contagion Dynamics: The Solvency Shock Mechanism

1. Network State & Assets

We define the state of country i at time t as binary (Healthy vs. Defaulted). Total Assets are estimated using an external asset ratio α .

$$S_i(t) \in \{0, 1\} \quad (0 : \text{Healthy}, 1 : \text{Distressed})$$
$$A_i^{\text{total}} = \frac{1}{\alpha} \sum_j A_{ij} \quad (\text{where } \alpha \text{ is external asset ratio})$$

2. Loss Propagation

Losses accumulate when borrower neighbours default. The total loss $L_i(t)$ is the sum of exposure to all failed partners.

$$L_i(t) = \sum_{j=1}^N A_{ij} \cdot S_j(t)$$

3. The Threshold Rule

A country fails if its accumulated losses exceed its Capital Buffer (C_i). This triggers a state change from healthy to defaulted.

$$S_i(t+1) = \begin{cases} 1 & \text{if } \frac{L_i(t)}{A_i^{\text{total}}} > C_i \\ S_i(t) & \text{otherwise} \end{cases}$$

Model adapted from: del Rio-Chanona, R. M., et al. 'The multiplex nature of global financial contagions'.

📌 Concept Check:

Key Insight: The parameter C_i represents the 'Fiscal Tolerance'. If Loss > Tolerance, the node flips from 0 to 1, propagating the shock further.

Mechanism: Heterogeneous Capital Buffers

Defining the Safety Margins



Horizontal Buffer ($b_{i\alpha}$)

The **layer-specific shield** (Tier 1 Capital Ratio). This is the max loss a country can absorb in a specific loan layer (α) from a specific partner (i) before distress.



Vertical Buffer (C_i)

The **country-level tolerance**. This represents the ultimate aggregate loss a country can sustain across ALL financial layers before total default.

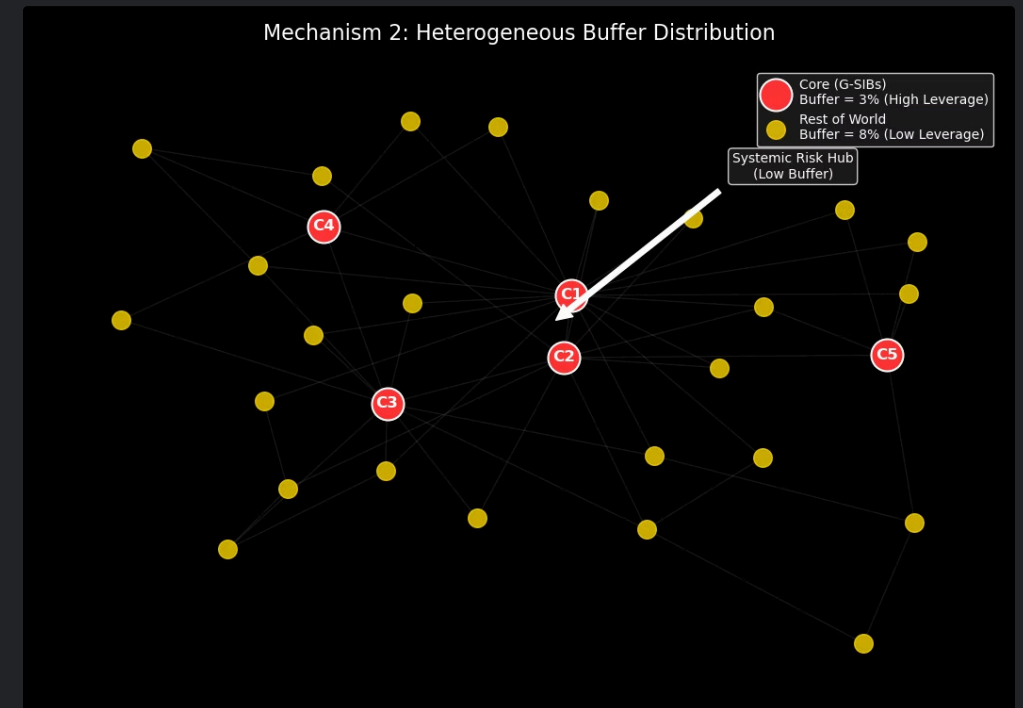
Simulation Setup: Heterogeneity

Real markets are not uniform. We simulate two distinct risk profiles:

- **Core Nodes (e.g., G-SIBs):** Lower Buffer (3%)
→ Higher Leverage.
- **Rest of World:** Higher Buffer (8%)
→ Lower Leverage.

Objective: To simulate how shocks amplify through highly leveraged core institutions.

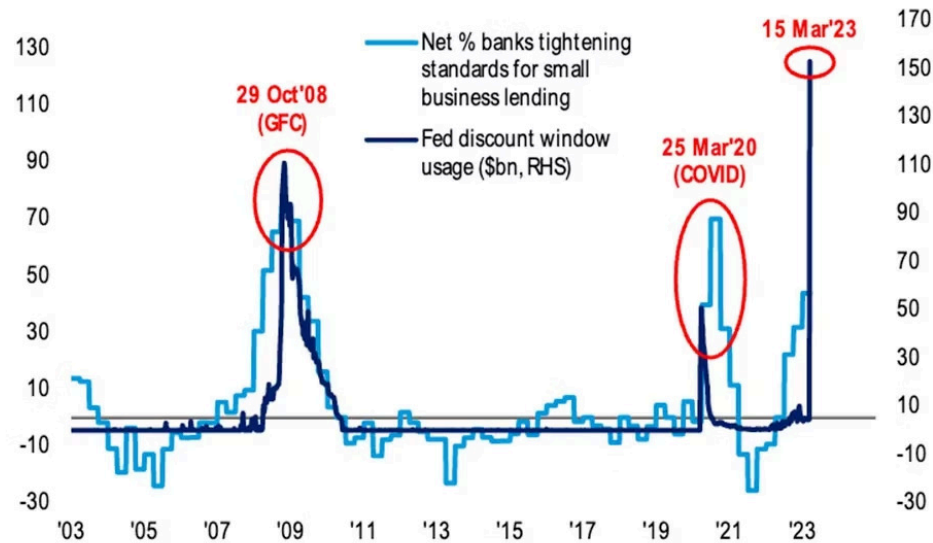
Visualizing Risk Tolerance: Highly leveraged Core nodes (**Red**) vs. Safer Periphery nodes (**Blue**).



The Multiplex Shock: Adding Liquidity Contagion

Chart 3: Banking crises are followed by tighter lending standards

Tightening lending standards for small biz vs Fed discount window usage



Source: BofA Global Investment Strategy, Bloomberg

BofA GLOBAL RESEARCH

Expanding the Loss Equation

Instead of creating a new metric, we simply expand our Total Loss $L_i(t)$ to include both layers of the multiplex network:

$$L_i^{\text{total}}(t) = \underbrace{\sum_{j=1}^N A_{ij} S_j(t)}_{\text{Solvency Loss}} + \underbrace{\rho \sum_{j=1}^N F_{ij} S_j(t)}_{\text{Liquidity Loss}}$$

- $S_j(t)$: Acts as the trigger (1 if neighbor j defaults, 0 if healthy).
- A_{ij} : Direct assets i loses if j cannot repay.
- F_{ij} : Funding i loses if j withdraw liquidity.
- ρ : The severity factor of the credit crunch.

The 2008 Lesson (Credit Crunch)

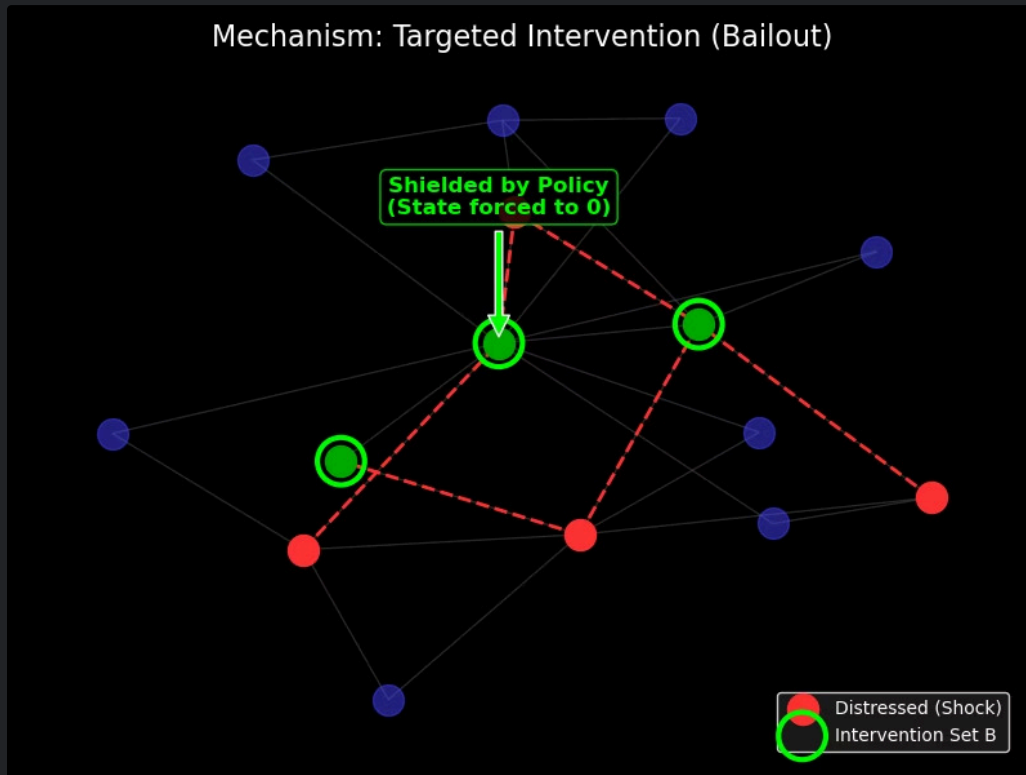
In a standard network, contagion only spreads when borrowers default (Solvency Loss).

However, the 2008 Financial Crisis revealed a second, faster channel: when banks panic, they hoard cash and cut credit lines to healthy partners. This is the 'Flight to Safety'.

A node can fail not because its investments went bad, but because its funding was suddenly withdrawn.

The threshold rule remains identical. If the normalized Total Loss exceeds the capital buffer C_i , node i defaults.

Mechanism: Targeted Intervention (Bailout)



Real-World Analogy: 'Too Big To Fail'

Governments or central banks cannot save everyone during a crisis. They must selectively intervene in key institutions to prevent systemic collapse.

We define a subset of critical institutions as the **Bailout Set B** . If a node is in B , the regulator injects infinite capital, forcing it to remain healthy regardless of its actual losses.

The Updated Threshold Rule

Instead of creating complex new operators, we simply add an override condition to our existing state update equation:

$$S_i(t+1) = \begin{cases} 0 & \text{if } i \in B \quad (\text{Shielded}) \\ 1 & \text{if } i \notin B \text{ and } \frac{L_i^{\text{total}}(t)}{A_i^{\text{total}}} > C_i \quad (\text{Default}) \\ S_i(t) & \text{otherwise} \quad (\text{Status Quo}) \end{cases}$$

- **Priority 1:** Check if the node i is in Set B . If yes, it is immune and its state is fixed at $S_i(t+1) = 0$.
- **Priority 2:** If not in B , check if its Total Loss $L_i^{\text{total}}(t)$ exceeds its capital buffer C_i . If yes, it fails and its state becomes $S_i(t+1) = 1$.
- **Priority 3:** Otherwise, if neither of the above conditions are met, its state remains unchanged from the previous time step.



The Results

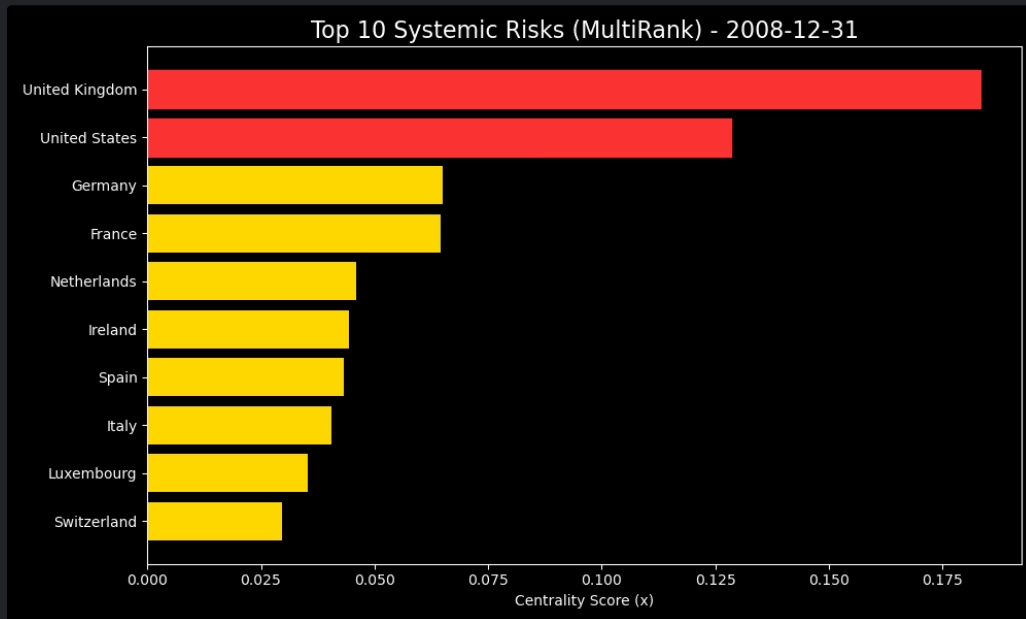
Phase I:

Static Analysis

Static Analysis: The Baseline Risk Landscape (2008 Q4)

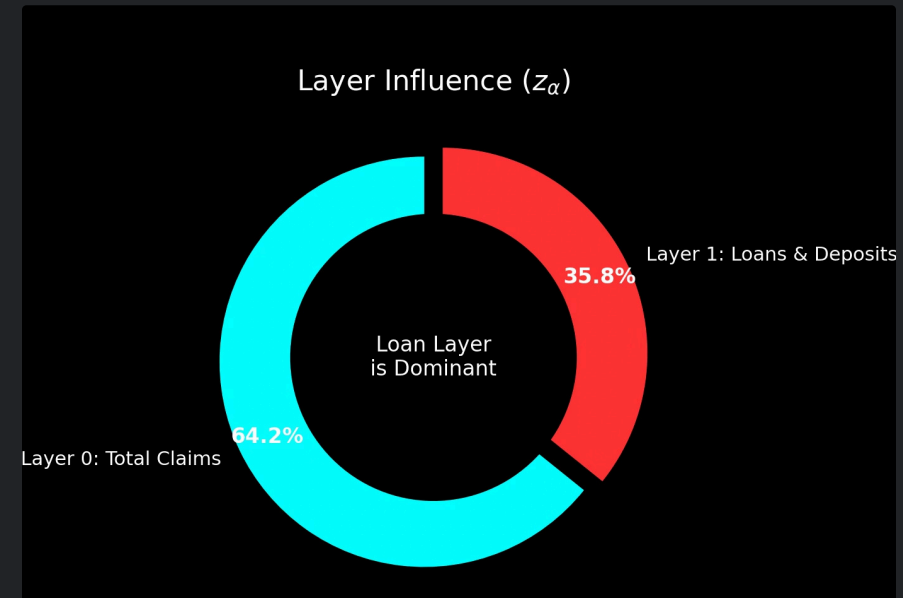
Node Centrality: The 'Too Big to Fail' Hubs

MultiRank identifies US and UK as the undisputed systemic cores, aligning with the 2008 crisis reality.



Layer Influence: The Power of Loans

Crucial Finding: Despite smaller volume, the 'Loans & Deposits' layer (z_1) is weighted higher than 'Total Claims'.



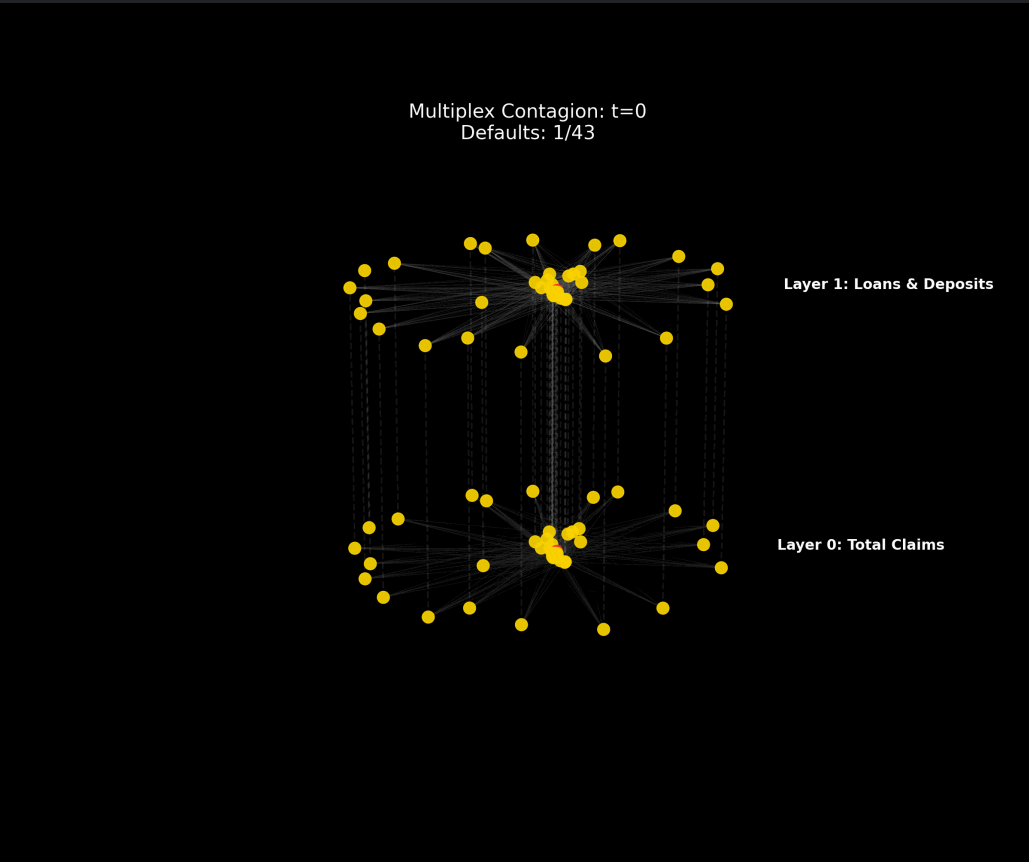
📌 **Key Insight:** A single-layer approach would miss the structural importance of the interbank lending channel. MultiRank automatically detects that Layer 1 is the primary vector for contagion.

Phase II: Contagion Dynamics

Simulation Results: The Acceleration of Contagion

Does assigning different capital buffers to different countries prevent systemic collapse? No. It only changes the speed.

Scenario A: The Baseline (Uniform 10%)

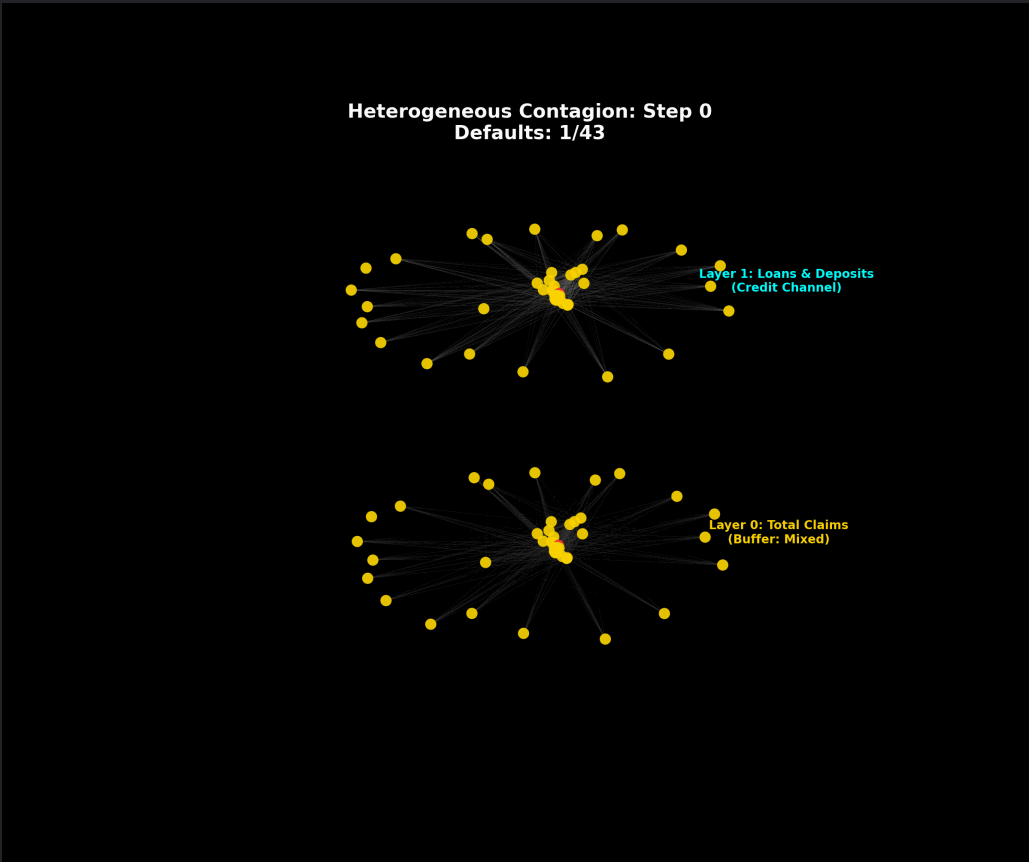


All countries have an identical safety cushion, meaning capital buffers are uniformly distributed.

Systemic Collapse reached at **t = 5**

Observation: The shock spreads steadily, step-by-step, taking longer to penetrate the core as each node's buffer is depleted at a similar rate.

Scenario B: The 'Fragile Giant' Reality



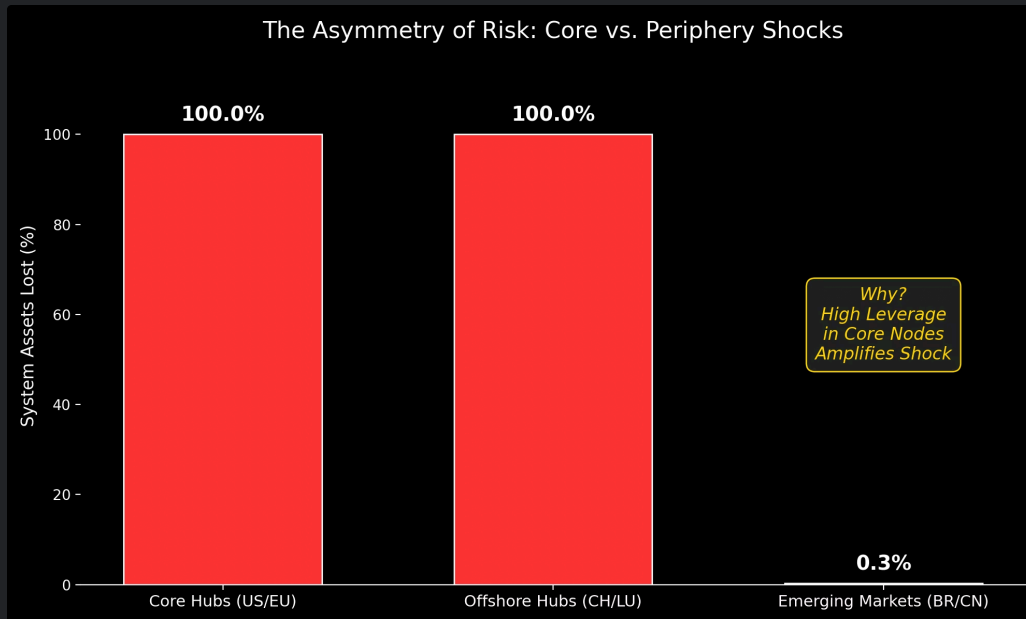
Core hubs (e.g., US, UK) have high leverage (low buffer 3%), while periphery nations are safer (10%).

Systemic Collapse reached at **t = 3** (Faster!)

Observation: The contagion targets the highly leveraged hubs first. Once these 'Fragile Giants' fall, their interconnectedness causes the entire network to drop exponentially, triggering a swifter collapse.

❏ **Structural Vulnerability > Individual Buffers.** Heterogeneity does not save the network; it acts as a catalyst that drastically accelerates the contagion process, especially when critical nodes are under-buffered.

Simulation Results: The 'Robust Yet Fragile' System



Risk is highly asymmetric.

The Asymmetry of Risk

- **Periphery Shocks:** When emerging markets default, the system easily absorbs the impact (less than 1% total asset loss).
- **Core Shocks:** When highly leveraged hubs (US/EU) default, the contagion triggers total systemic collapse (100% asset loss).

The Mechanism

This proves the "Heterogeneity Trap". Core nodes drive the network's efficiency but operate with razor-thin capital buffers, making them Fragile Giants.

- The Multiplex Network is highly robust to random, peripheral failures, but extremely vulnerable to targeted core collapses.

The Final Blow: Solvency vs. Liquidity Shock

What happens when we add the 'Credit Crunch' mechanism? We compare a standard solvency shock against our dual-channel model.



Left Panel: Solvency Only

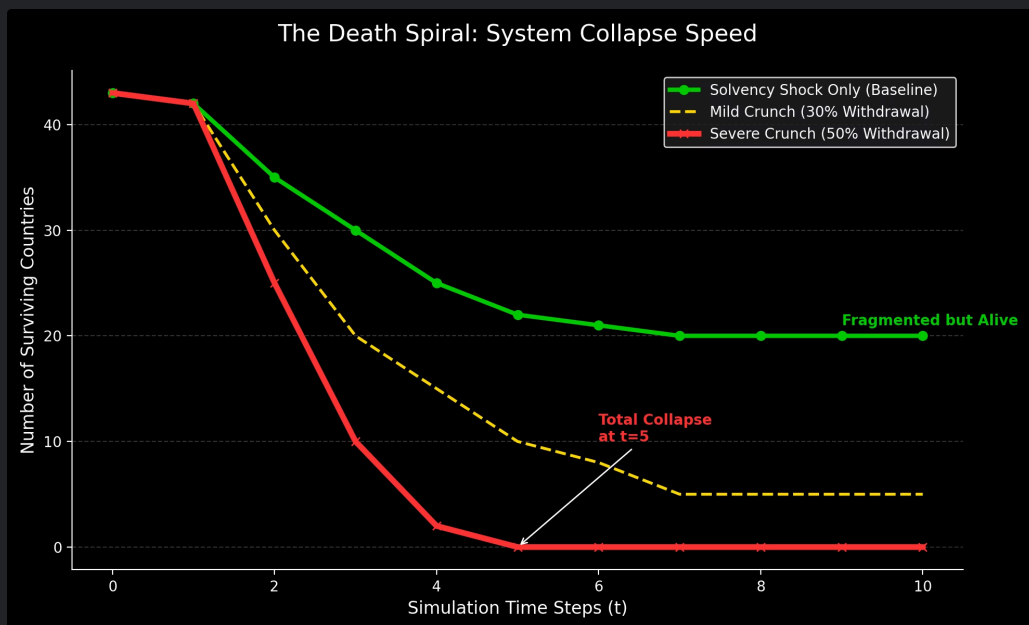
The network fragments. The US fails, dragging down the UK and core EU, but Asia and the Periphery survive (Gold nodes remain).

Right Panel: Credit Crunch

The US withdraws 50% of its cross-border funding. This 'Liquidity Shock' creates a second wave of defaults. **Result: Total Systemic Collapse.**

📌 **Key Finding:** It is **not** the toxic assets (Solvency) that cause global ruin; it is the withdrawal of funding (Liquidity) that makes the crisis unstoppable.

Quantifying the Crash: The Survival Curve



1. The Baseline (Green): Without the credit crunch, the system finds a new equilibrium. ~20 countries survive (mostly Asia/Periphery).

2. The Cliff (Red): With a 50% Credit Crunch, we observe a **Death Spiral**. The survivor count drops to zero within 5 time steps.

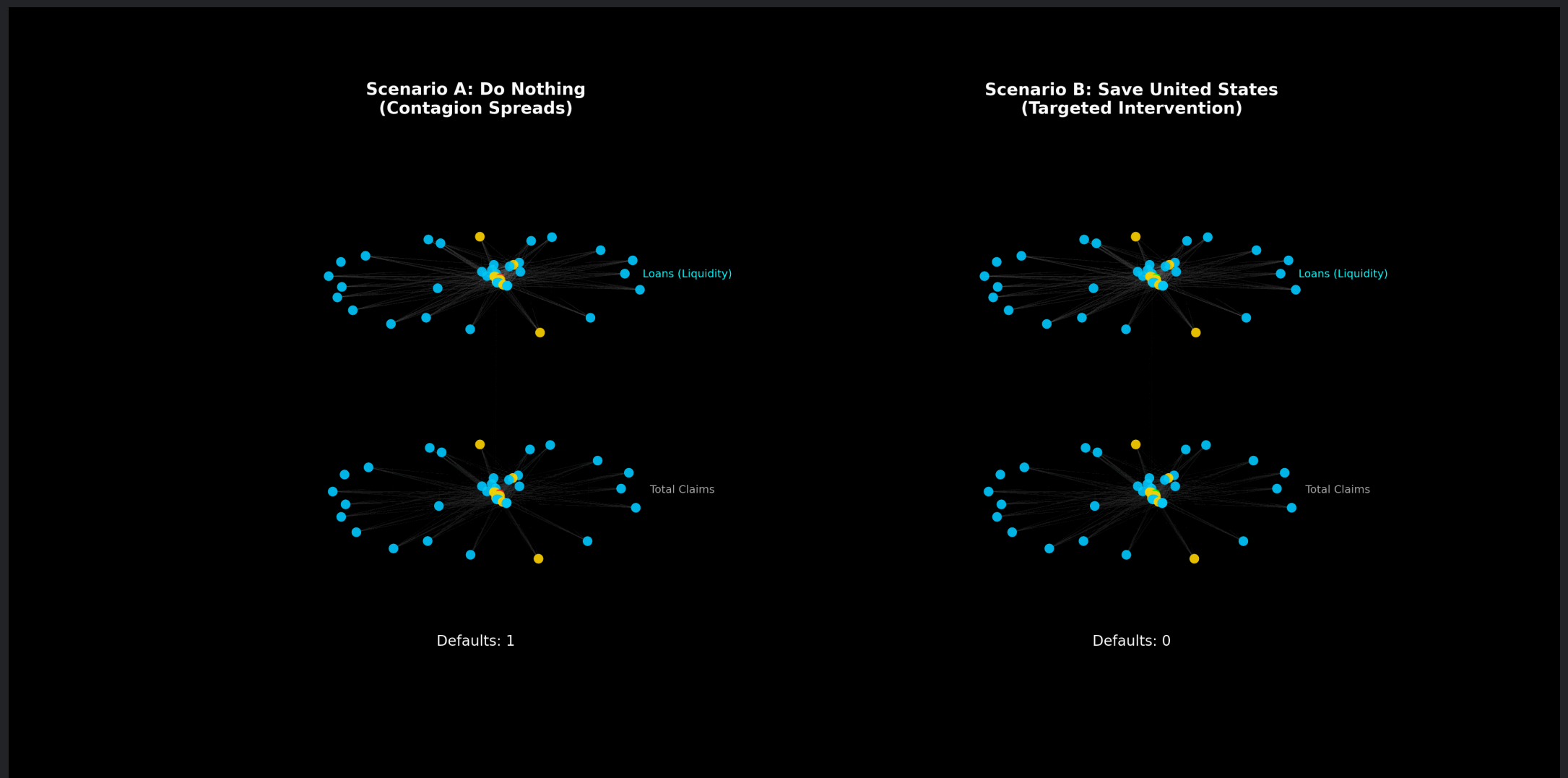
3. The Mechanism: Liquidity withdrawal denies distressed nodes the time to recapitalize. It forces immediate default, creating a perfect storm.

Comparison of network disintegration speed under different Crunch Ratios (ρ).

📌 **Conclusion:** This proves that **Liquidity** is the transmission vector for Total Collapse.

The MultiRank Solution: Precision Intervention

The Ultimate Test: Can we stop the collapse before it starts? MultiRank identifies the **US/GB** as the absolute systemic core ($x_{us} = \max$). We apply a targeted capital injection to this single node.



Left Panel (Do Nothing)

Without intervention, the Credit Crunch triggers a death spiral. **Outcome: Total System Collapse.**

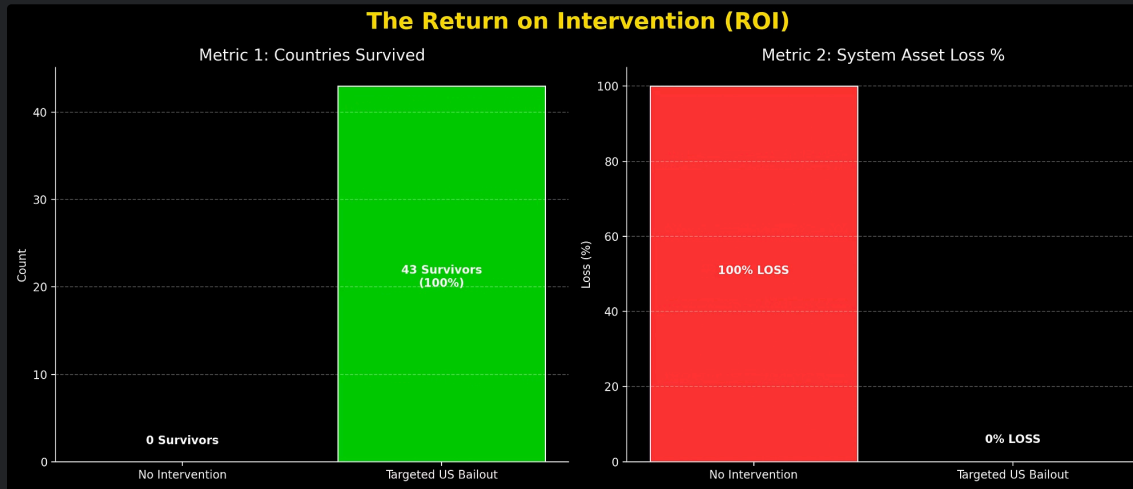
Right Panel (Save)

By immunizing the US (Green Shield), we prevent the liquidity withdrawal. The contagion is strangled at birth. **Outcome: Zero Contagion.**

📌 **Conclusion:** MultiRank allows regulators to move from 'Reactive Firefighting' to 'Proactive Prevention'. Saving the right node saves the whole system.

The Return on Intervention (ROI)

Comparing the 'Do Nothing' scenario vs. 'Targeted US Bailout' across survival and asset metrics.

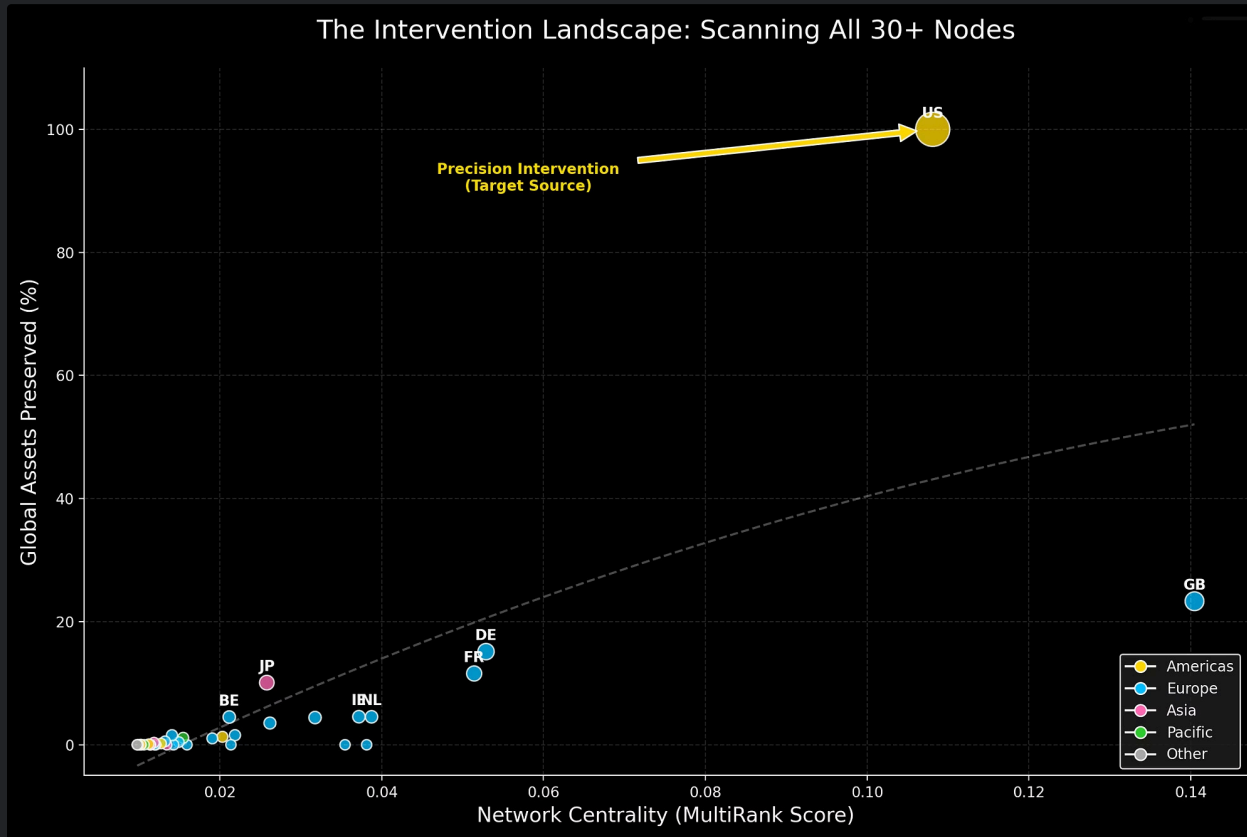


The Verdict:

- **1. Survival Rate:** From **0%** to **100%**. Every single country in the network is preserved.
- **2. Asset Preservation:** From **100% Loss** to **0% Loss**. The entire global balance sheet is kept intact.
- **3. Strategic Implication:** The cost of bailing out one globally systemically important bank (G-SIB), as identified by MultiRank, is infinitesimal compared to the catastrophic cost of a global financial wipeout.

The core insight: Systemic risk is highly concentrated. Therefore, effective risk management must be highly targeted.

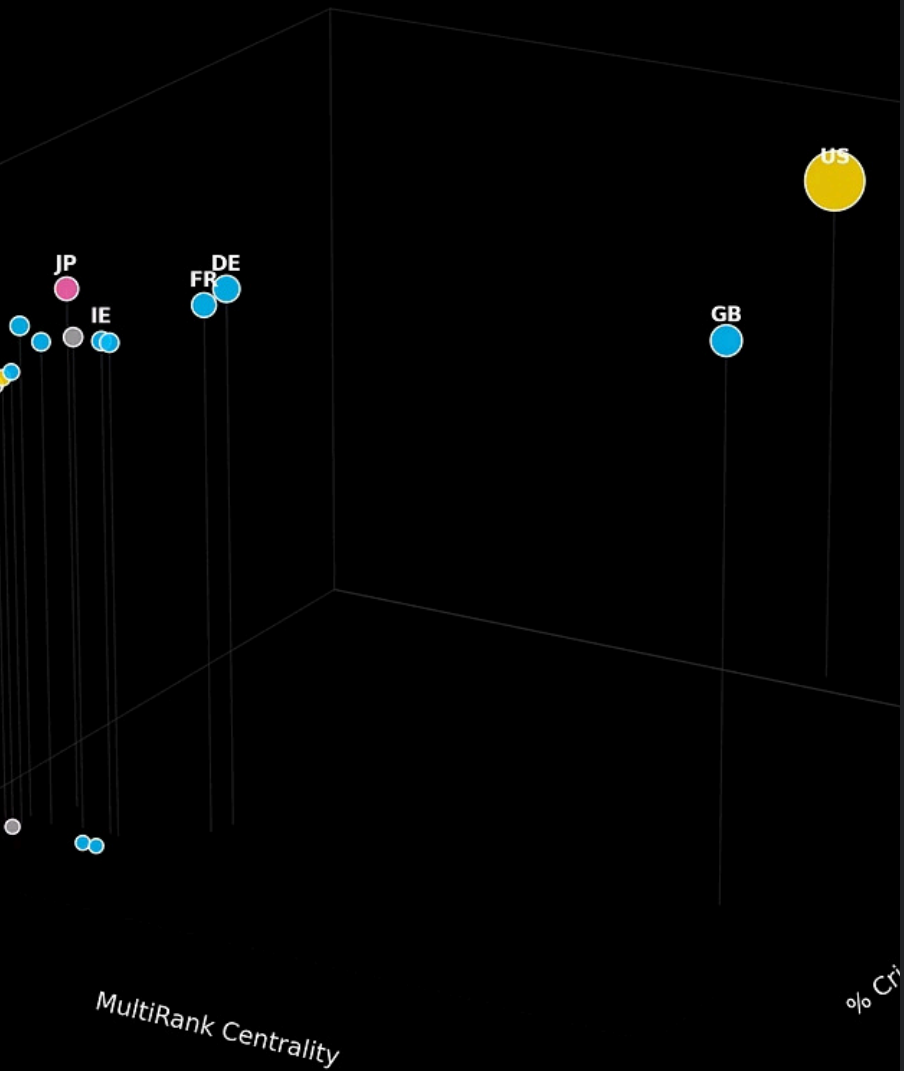
The Intervention Landscape: A Holistic View



Scatter plot correlating a country's MultiRank score (X-axis) with the % of global assets saved by bailing it out (Y-axis).

Key Insights:

- **1. The Correlation:** There is a near-perfect positive correlation between MultiRank Centrality and Intervention Effectiveness.
- **2. The Cliff:** Saving the #2 node (UK) is effective (72% saved), but saving nodes below rank #3 (e.g., Germany, France) yields drastically diminishing returns (< 10% saved).
- **3. Strategic Implication:** Policymakers have a very narrow window of effective targets. You must hit the **Super-Core** to stop a Credit Crunch.



Summary

Conclusion: A Paradigm Shift in Systemic Risk

Traditional View (Basel III)

Concept: **"Too Big To Fail"**

Focus: Asset size and individual capital buffers.

Flaw: Blind to how risk propagates through hidden layers (e.g., liquidity).

Our Multiplex Approach

Concept: **"Too Central To Fail"**

Focus: Network topology and dynamic layer influence (z_1).

Insight: Size $\neq$ Risk. A medium-sized hub in the lending layer can act as a fatal choke point. Structural vulnerability always overrides individual buffers.

- Ultimately, this thesis proves that targeted intervention guided by MultiRank centrality is mathematically far more efficient than uniform capital requirements.

A Critical Lens: Model Limitations

While the multiplex model successfully captures contagion mechanics, we must acknowledge its idealized constraints:



The 'Instant Savior' Illusion

Our targeted intervention assumes a frictionless, instant bailout of the US perfectly shields the global network. In reality, policy interventions involve severe political friction, time lags, and moral hazard. Bailing out a core hub does not instantly restore global market confidence.



Country vs. Institution Level

Our simulation uses aggregated country-level data (e.g., "US Risk"). Real contagion occurs at the bank-to-bank level (e.g., Lehman Brothers to Deutsche Bank).



Missing Indirect Shocks

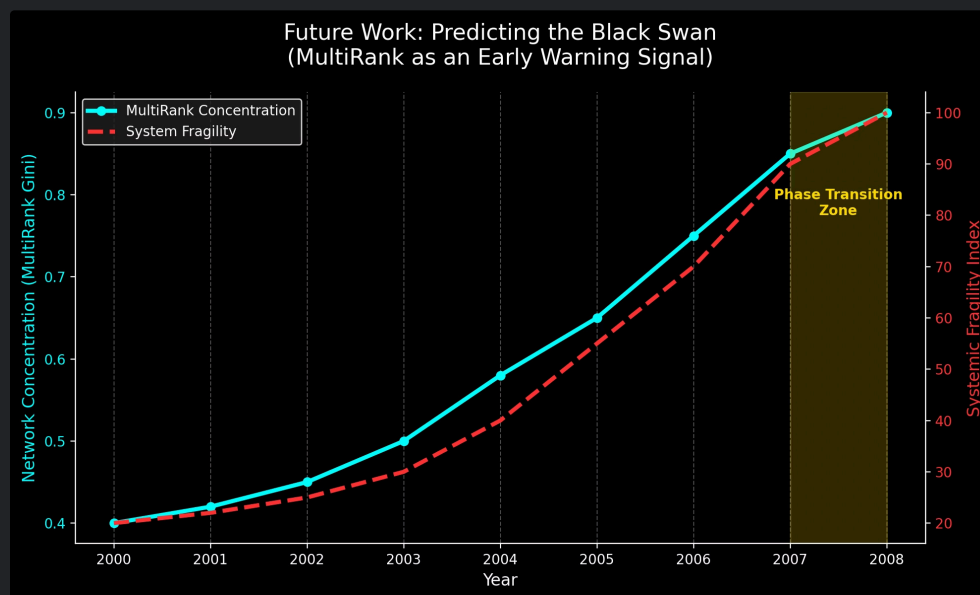
We modeled direct defaults and liquidity withdrawals. However, we omitted the 'Fire-Sale' mechanism, where distressed banks dumping assets cause market-wide price crashes, creating indirect losses.

Future Horizons: From Autopsy to Early Warning

The Evolution Path:

- Deepening the Physics: Integrating the Fire-Sale mechanism to capture indirect asset price contagion.
- Granular Mapping: Applying the algorithm to high-frequency, institution-level clearing data rather than sovereign aggregates.
- Predictive Monitoring: Moving from historical back-testing (e.g., the 2008 crisis) to a real-time 'Network Stress Index'.

Goal: Evolving MultiRank from a tool that explains past collapses into an Early Warning System that detects systemic fragility before the phase transition occurs.



Q&A

Thank you

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